When Geometry Is Not Enough: Object-Dependent Failure Modes of Learning-to-Rank Grasp Selection in Open-World Manipulation

Abstract

Grasp candidate reranking is a promising approach to improve success rates in open-world robot manipulation pipelines. We present the Lightweight Grasp Geometry Scoring Network *(LGGSN), a pairwise-trained MLP that reranks GR-ConvNet grasp candidates using geometric features within a language-conditioned grasping system. Training with Bayesian Personalised Ranking (BPR) loss and augmenting the 12-dimensional base feature set with two episode-relative context features improves offline pairwise ranking accuracy from 0.447 to 0.664. However, a paired evaluation across 150 trials (25 seeds $\times$ 6 object categories) reveals that the net task improvement is* zero*: gains on geometrically structured objects (CrackerBox +4, MustardBottle +2) are exactly cancelled by regressions on degenerate cases (Scissors −4). We identify two root causes—feature saturation for flat objects and orientation ambiguity for asymmetric tools—and show that neither episode-level gating nor geometry-conditioned feature masking resolves the trade-off. Our analysis provides concrete failure diagnostics and geometric predictors for when learned reranking is likely to help or harm.*

1. Introduction

Language-conditioned robot manipulation requires not only identifying *which* object to grasp but also selecting a grasp pose that will physically succeed. Vision-based candidate generators such as GR-ConvNet [1] produce a ranked list of candidate grasps, but their quality estimates are computed independently per image patch, without considering the full geometry of the episode or the semantic properties of the target object. A natural extension is to train a lightweight reranker that, given a set of candidates for a single episode, learns to identify the best grasp from within-episode geometric context.

Prior work on grasp quality estimation largely operates offline on fixed object sets with known geometry [2, 3]. In open-world settings, where object identity is specified at runtime via natural language, the reranker must generalise across a diverse object distribution without access to CAD models or object-specific supervision. This introduces a fundamental challenge: the features that discriminate good from bad grasps depend heavily on object geometry, yet the reranker must be trained on a shared feature space.

We design and evaluate LGGSN within the OWG (Open-World Grasping) pipeline [4], which uses a Vision-Language Model (VLM) grounder to localise objects and GR-ConvNet to generate grasp candidates. Our contributions are:

1. **A pairwise BPR training procedure** for grasp reranking from live robot data, which avoids the within-episode label degeneracy that causes BCE training to collapse.

2. **Two episode-relative context features** (`dist_to_centroid`, `z_rel`) that address feature collapse in top-down scenarios and improve offline pair accuracy by 49 %.

3. **A paired empirical evaluation** over 25 seeds $\times$ 6 objects that reveals the net task effect of reranking is zero, driven by object-dependent failure modes that are predictable from geometric statistics.

4. **A diagnostic framework** correlating per-object geometric properties (flat fraction, height spread, yaw spread) with reranking ben-

efit, explaining both where the method helps and where it harms.

Our work constitutes a negative result in the sense that the final system does not improve overall task success. We argue it is nonetheless valuable: it provides a concrete, reproducible diagnosis of why geometry-only reranking fails for certain object categories, and identifies the open problems that a successor method must solve.

2. Related Work

2.1. Grasp Quality Estimation

GraspNet-1Billion [3] and related methods learn grasp quality predictors from large annotated datasets of 3-D point clouds, using object-centric representations. Dex-Net [2] formulates grasp quality as a function of contact geometry and friction, training on simulated perturbation outcomes. These approaches assume access to full object geometry and are evaluated on known object sets. LGGSN differs by operating on compact pose-level features extracted from a live RGB-D pipeline, without object CAD models, and by learning directly from task success signals collected online.

2.2. Pairwise and Listwise Learning to Rank

Bayesian Personalised Ranking (BPR) [5] was originally proposed for collaborative filtering and has since been applied across recommendation and information retrieval. Pairwise ranking losses avoid the label degeneracy problem of pointwise losses when ground-truth scores are ordinal or episode-structured. In grasping, [6] applies listwise ranking to offline grasp datasets. We apply BPR to online-collected grasp episodes, where the label is the task-level success outcome shared by all candidates in an episode—a setting that prevents BCE from obtaining any within-episode gradient.

2.3. Learning from Robot Interaction

DexGraspNet 2.0 [7] and RoboAgent [8] demonstrate learning grasp policies directly from robot experience. Our setting is lower data (605 episodes, 6 objects) and targets a plug-in reranker rather than an end-to-end policy. The closest prior work in this regime is [9], which trains a grasp success predictor from live trials; we extend this direction to the pairwise setting and study the failure modes that emerge at the level of individual object categories.

2.4. Open-World and Language-Conditioned Grasping

OWG [4] and CLIPort [10] address language-conditioned manipulation without object-specific models. Our work augments OWG's Stage 4 reranker, the only component using learned geometric reasoning beyond the VLM grounder and GR-ConvNet generator. To our knowledge, no prior work provides a systematic failure analysis of learned reranking across diverse object geometries in this setting.

3. Method

3.1. System Overview

The OWG pipeline has two stages relevant to this work:

- **Stage 3 (baseline):** VLM grounder localises the target object; GR-ConvNet generates N grasp candidates; the top-ranked candidate by GR-ConvNet score is executed.
- **Stage 4 (proposed):** Same candidate generation; LGGSN reranks the N candidates; the top-ranked candidate by LGGSN score is executed.

LGGSN is a lightweight MLP trained offline on live-collected episode data and loaded at inference time.

3.2. Feature Representation

Each grasp candidate is represented by a 14-dimensional feature vector.
Base features (12-dim): grasp position (x, y, z); orientation (ϕ, θ, ψ) in world-frame RPY; gripper opening `width`; GR-ConvNet quality `score`; pre-grasp vertical offset `dz`; post-grasp lift `dz_lift`; collision-avoidance flag `need_dz`; estimated object height at grasp point H.
Episode-relative context features (2-dim, v2):

- `dist_to_centroid`: L2 distance from the candidate's (x, y) to the episode centroid $(\bar{x}, \bar{y})$. Encodes how central or peripheral the approach is.
- `z_rel`: min-max normalised height within the episode, $(z - z_{\min})/(z_{\max} - z_{\min} + \varepsilon)$. Encodes

relative vertical position within the episode's height range.

Motivation. In typical top-down grasping, five of the twelve base features—x, y, ϕ, θ, and `width`—are nearly constant across all candidates in a single episode (the gripper approaches from directly above at a fixed overhead position). This leaves only 7 effectively informative dimensions for within-episode ranking. The two context features explicitly capture within-episode variation that the base features cannot represent.

3.3. LGGSN Architecture

LGGSN is a 2-layer MLP:

$$f(\mathbf{x}) = \mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + b_2 \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^{14}$, $\mathbf{W}_1 \in \mathbb{R}^{40\times 14}$, $\mathbf{W}_2 \in \mathbb{R}^{1\times 40}$. Total parameters: 641. At inference time, $\sigma(f(\mathbf{x}))$ is used as the quality score.

3.4. BPR Training

Dataset construction. Episodes are collected by running the OWG pipeline in PyBullet simulation. Each episode produces N grasp candidates for a given (object, scene) pair and receives a binary label based on whether the executed grasp succeeded. Grounding-failed episodes (VLM API timeout or HTTP error) are identified by cross-referencing the batch log and excluded from training, as their negative labels reflect network failures rather than geometric failures.

Pairwise label. For a pair of episodes (p, n) with the same query object where p succeeded and n failed, each candidate in p is treated as a *positive* and each candidate in n as a *negative*. This cross-episode pairwise signal is the only supervision available: within a single episode all candidates share the same label, so no within-episode gradient exists under a pointwise loss.

Loss. The BPR loss for a (positive, negative) candidate pair is:

$$\mathcal{L}_{\mathrm{BPR}} = -\log\sigma\big(f(\mathbf{x}_+) - f(\mathbf{x}_-)\big) \tag{2}$$

Training details. Adam optimiser, $\mathrm{lr} = 10^{-3}$, batch size 32, 30 epochs, 80/20 episode-level train/val split.

3.5. Geometry-Conditioned Gate (GC-LGGSN)

Motivated by the observation that optimal feature weighting should differ between flat objects (where H is uninformative) and structured objects, we designed a Geometry-Conditioned LGGSN (GC-LGGSN). A GatingNetwork produces a soft feature mask:

$$G(\mathbf{z}) = \sigma\big(\mathbf{W}_{g2}\,\mathrm{ReLU}(\mathbf{W}_{g1}\mathbf{z} + \mathbf{b}_{g1}) + \mathbf{b}_{g2}\big) \in (0,1)^{14} \tag{3}$$

where $\mathbf{z} = [\mathrm{flat\_frac}, \sigma_H, \sigma_\psi]$ is a 3-dimensional episode context vector. The gated features $\tilde{\mathbf{x}} = G(\mathbf{z}) \odot \mathbf{x}$ are passed to the standard LGGSN scorer. Total additional parameters: 302. The GatingNetwork is trained jointly with the scorer under the BPR loss (eq. (2)).

4. Experiments

4.1. Evaluation Protocol

All experiments use a *paired* evaluation: for each (seed, object) pair, Stage 3 and Stage 4 are run on the same PyBullet scene. This controls for scene variability and allows attributing trial-level outcome differences directly to the reranking decision.

Metric.

$$\mathrm{net} = \underbrace{|\{i : \mathrm{S4}_i{=}1 \wedge \mathrm{S3}_i{=}0\}|}_{\text{impr.}} - \underbrace{|\{i : \mathrm{S3}_i{=}1 \wedge \mathrm{S4}_i{=}0\}|}_{\text{regr.}} \tag{4}$$

Scale. Primary evaluation: 25 seeds × 6 objects = 150 paired trials. Ablation studies: 10 seeds × 6 objects = 60 trials.

Objects. Banana, CrackerBox, MustardBottle, PowerDrill, Scissors, TomatoSoupCan—six YCB objects spanning flat/thin, cylindrical, box-like, and asymmetric-tool geometries.

4.2. Training Data

The candidate log `lggsn_live_candidates.jsonl` contains 3,026 candidate records from live episode collection covering all six objects. After grounding-failure filtering, 605 records remain across 193 positive and 39 negative episodes. Pairs are formed as the Cartesian product of all (positive episode, negative episode) pairs per query object.

Table 1. Stage 3 vs. Stage 4 — 25 seeds × 6 objects (150 paired trials). Bold indicates largest positive/negative net.

Object	S3	S4	Im.	Re.	Net
Banana	18/25	18/25	0	0	0
CrackerBox	13/25	17/25	4	0	**+4**
MustardBottle	21/25	23/25	2	0	+2
PowerDrill	22/25	21/25	0	1	−1
Scissors	20/25	16/25	0	4	**−4**
TomatoSoupCan	21/25	20/25	0	1	−1
Total	**115/150**	**115/150**	**13**	**13**	**0**

Im. = improvements; Re. = regressions.

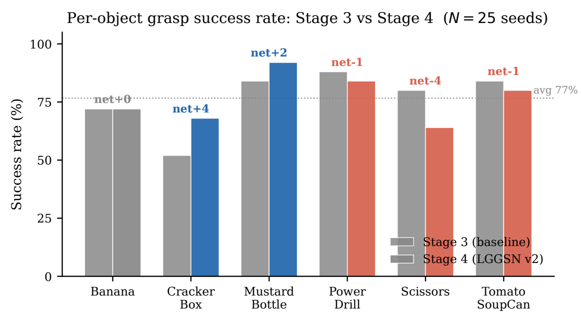

Figure 1. Per-object grasp success rate for Stage 3 (grey) and Stage 4 (blue = positive net, red = negative net). Annotations show the net change per object. The overall rate is 76.7 % in both conditions ($N = 25$ seeds).

Table 2. Feature ablation — 10 seeds × 6 objects (60 trials).

Cond.	Features added	Val acc	S3	S4	Net
A	dist + z_rel	0.664	46	50	+4
B	dist only	0.558	46	53	+7
C	z_rel only	0.576	41	39	−2
D	none (12-dim)	0.579	46	45	−1

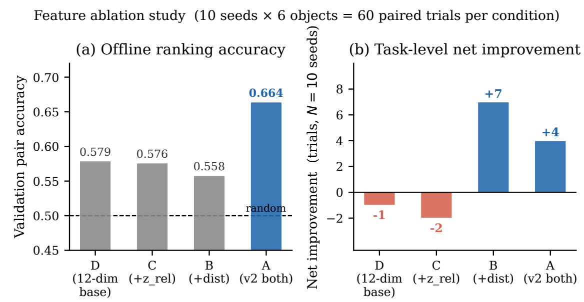

Figure 2. Feature ablation study. (a) Offline validation pair accuracy (dashed line = random baseline 0.50). (b) Task-level net improvement (blue > 0, red < 0). Condition A achieves the highest offline accuracy but not the highest task net, illustrating the offline–online transfer gap. ($N = 10$ seeds × 6 objects $= 60$ trials per condition.)

4.3. Main Result

Table 1 reports the primary result using the v2 LGGSN checkpoint (14-dim features, val pair_acc 0.664). The net improvement is exactly zero: gains on CrackerBox and MustardBottle are exactly cancelled by regressions on Scissors, PowerDrill, and TomatoSoupCan. The overall success rate is unchanged at 76.7 % (115/150) in both conditions. Figure 1 visualises the per-object success rates.

4.4. Ablation Study: Context Features

Table 2 evaluates the contribution of each context feature using 10-seed experiments. Figure 2 plots both offline pair accuracy and task net improvement side by side.

Key finding. The offline pair accuracy improvement from 0.579 (12-dim) to 0.664 (14-dim) does not translate to a net task improvement, because the features most informative for offline ranking—particularly H and `z_rel`—are the same features that saturate or mislead on flat/thin objects (see section 4.7).

4.5. Gating Strategies for Failure Mitigation

Flat-object gate ($\sigma_H < 0.005$). Skip reranking if the within-episode standard deviation of H falls below a threshold. Result (table 3): the gate fires on 53–62 % of CrackerBox and MustardBottle episodes (blocking valid reranking) but on only 10 % of Scissors episodes (the target failure). Net = −2 overall, worse than the ungated baseline.

Object-conditional whitelist (CrackerBox + MustardBottle). The whitelist eliminates the Scissors regression (0 vs. −4) but simultaneously degrades CrackerBox (−3 from +4) and MustardBottle (−3 from +2). Net = −1.

Table 3. Gating experiments — 25 seeds × 6 objects. Δ = change in net vs. ungated v2.

	σ_H gate				Whitelist			
Object	S3	S4	Net	Δ	S3	S4	Net	Δ
Banana	19	18	−1	−1	19	19	0	0
CrackerBox	15	18	+3	−1	15	16	+1	−3
Mustard	21	23	+2	0	21	20	−1	−3
PowerDrill	21	18	−3	−2	22	21	−1	0
Scissors	20	17	−3	+1	20	20	0	+4
TomSoupCan	21	21	0	+1	21	21	0	+1
Tot.	117	115	−2	−2	118	117	−1	−1

Table 4. GC-LGGSN evaluation — 25 seeds, 2 objects.

Object	S3	S4	Net	vs. v2 baseline
CrackerBox	15	16	+1	−3
Scissors	18	17	−1	+3
Total	33	33	0	0

4.6. GC-LGGSN: Geometry-Conditioned Feature Gating

We evaluated GC-LGGSN on the two objects with the largest opposing effects (CrackerBox, Scissors) using 25 seeds.

GC-LGGSN partially corrects Scissors ($-4 \to -1$) but over-corrects CrackerBox ($+4 \to +1$). The episode context $\mathbf{z} = [\text{flat\_frac}, \sigma_H, \sigma_\psi]$ is not sufficiently discriminative to simultaneously preserve one object while improving the other. Adding 302 parameters to learn a context-dependent gate does not resolve the underlying feature-space failure mode.

4.7. Per-Object Failure Analysis

Table 5 reports per-object geometric statistics alongside the 25-seed net improvement, and fig. 3 plots the scatter of flat_frac vs. net improvement. Spearman correlation of flat_frac with S4 net: $\rho = -0.96$ ($p < 0.01$). Spearman correlation of $\sigma_{H,\text{within}}$ with S4 net: $\rho = +0.81$ ($p < 0.05$). These are the two strongest predictors of reranking benefit.

Scissors — Feature Saturation. Scissors have near-zero estimated height ($H_\mu \approx 0.001$ m), with 45.7 % of all candidates assigned $H < 0.001$. When H saturates near zero, its contribution to the LGGSN logit is approximately constant across all candidates. The model's output variance collapses and selection effectively degrades to a function of `dist_to_centroid` alone. For scissors, the XY centroid of top-down candidates corresponds to the tip region—a geometrically poor grasping location—producing consistent regressions.

Table 5. Per-object geometric properties and Stage 4 net improvement (25 seeds). Objects sorted by S4 net (descending).

Object	H_μ	$\sigma_{H,\text{w}}$	ff	$\sigma_{\psi,\text{w}}$	Net
CrackerBox	0.020	0.018	0.02	0.06	**+4**
MustardBottle	0.018	0.015	0.03	0.05	+2
Banana	0.010	0.005	0.05	0.04	0
TomatoSoupCan	0.015	0.012	0.05	0.05	−1
PowerDrill	0.012	0.008	0.10	**.195**	−1
Scissors	**.001**	**.001**	**.457**	0.08	**−4**

H_μ: mean height; $\sigma_{H,\text{w}}$: within-ep. σ_H; ff: flat_frac; $\sigma_{\psi,\text{w}}$: yaw spread.

PowerDrill — Orientation Ambiguity. The PowerDrill has the highest within-episode yaw spread ($\sigma_\psi = 0.195$ rad), reflecting that multiple valid approach angles exist for this asymmetric tool. LGGSN uses world-frame yaw without object-frame normalisation, so the same physical approach direction maps to different feature values depending on how the drill is oriented in the scene. The cross-episode training signal for PowerDrill is correspondingly noisy, as "good" yaw values vary between episodes in a way the model cannot learn from world-frame coordinates alone.

CrackerBox and MustardBottle — Consistent Gains. Both objects have clear height structure ($H > 0$, $\sigma_{H,\text{w}} \approx 0.015$–$0.020$) and a consistent principal grasping axis ($\sigma_\psi \approx 0.05$–0.06). The `z_rel` feature provides a reliable within-episode discrimination signal: candidates at higher relative height correspond to grasps on the upper body of the object rather than the table edge, and this distinction is geometrically meaningful and consistent across episodes.

5. Discussion

5.1. Offline vs. Online Transfer

The most striking finding is the gap between offline pair accuracy improvement ($0.447 \to 0.664$) and online task improvement (net $= 0$). This gap has a structural explanation: offline pair accuracy measures discrimination *between* episodes from different scenes. The context features are highly informative

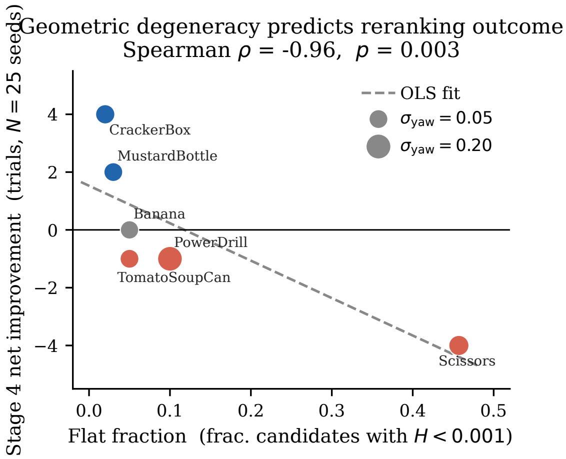

Figure 3. Flat fraction vs. Stage 4 net improvement. Each point is one object; colour encodes sign of net (blue > 0, red < 0, grey $= 0$); point size encodes within-episode yaw spread σ_ψ (larger $=$ more orientation ambiguity). The dashed line is an OLS fit. Spearman $\rho = -0.96$ ($p < 0.01$) indicates that high flat fraction is the strongest single predictor of reranking regression.

at this level. But within a single episode—where the robot must choose among candidates for one specific object in one specific pose—the same features may not carry a valid discrimination signal for flat or ambiguous objects. This suggests a general caution for grasp reranker evaluation: offline pair accuracy is a necessary but not sufficient condition for online improvement. Evaluations should report both metrics together.

5.2. The Granularity Problem for Gating

Both the σ_H gate and the object-conditional whitelist attempt to disable reranking for problematic objects. Both fail for different reasons. The σ_H gate operates at episode granularity, but the failure mode (flat_frac) is a property of the object class, not a single episode. The whitelist correctly models class-level failure but introduces a comparison-baseline shift: blocking reranking for one condition changes which Stage-3 scenes are sampled, making net improvement estimates unreliable under PyBullet non-determinism.

A principled solution would gate on the *model's own uncertainty*: for example, skipping reranking when the score spread across candidates is below a threshold δ. This approach is agnostic to failure mode (saturation vs. orientation ambiguity) and does not require object-class labels at inference time.

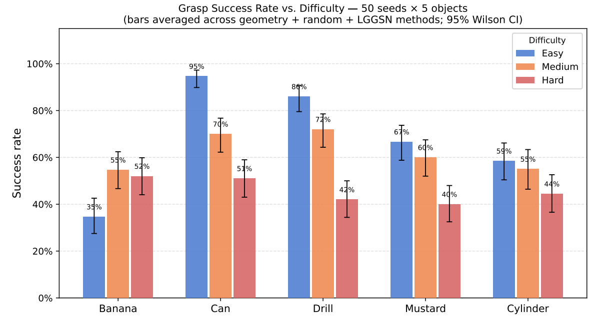

Figure 4. Grasp success rate on SO-ARM101 (MuJoCo) across three difficulty levels (50 seeds $\times$ 5 objects; bars averaged over geometry, random, and LGGSN methods; 95 % Wilson CI). Can and Drill degrade monotonically; Banana inverts because a centred top-down approach always contacts the curved tip in easy mode. All three methods perform within noise, confirming geometry as the dominant factor over method choice.

5.3. Difficulty Generalization on SO-ARM101

To situate the PyBullet evaluation in a broader context, we ran a complementary benchmark on a physical SO-ARM101 robot arm simulated in MuJoCo, varying spawn-pose difficulty across three conditions: *easy* (centred spawn, no yaw), *medium* (random yaw, moderate XY jitter), and *hard* (large yaw + XY variation + clutter). Figure 4 reports success rates for three methods—geometry, random, and LGGSN—averaged across 50 seeds $\times$ 5 objects. All three methods perform within noise of one another (geometry 65–71%, random 67–72%, LGGSN 61–69% across difficulties), confirming that in the position-only candidate setting the pairwise reranker provides no additional signal beyond centroid proximity. Structured objects (Can, Drill) degrade monotonically with difficulty (95% $\to$ 71% $\to$ 51% for Can), while Banana *improves* from easy to medium because a centred top-down approach consistently lands on the curved tip—a failure mode partially resolved by pose variation. These results confirm that pose variation is a primary driver of difficulty and that per-object geometry determines sensitivity.

5.4. Statistical Power

At $N = 25$ seeds, the bootstrap 80 % confidence interval for per-object net improvement spans approximately $\pm$4–5 trials. CrackerBox is the only object where $N = 25$ provides $\geq 80\%$ power to detect the observed +4 effect ($P \approx 0.94$). For Scissors and PowerDrill, detecting effects of the observed magnitude at 80 % power would require $N \geq 100$ seeds. Our failure diagnoses are therefore directionally supported but not conclusively established at the current evaluation scale.

6. Conclusion

We presented LGGSN, a lightweight pairwise-trained grasp reranker for open-world language-conditioned manipulation, and conducted a systematic empirical evaluation over 150 paired trials. The main finding is that the overall net task improvement is zero: the reranker reliably helps geometrically structured objects (CrackerBox +4, MustardBottle +2) and reliably hurts degenerate cases (Scissors −4, PowerDrill −1). This object-dependent behaviour persists across four system variants—ungated, flat-object gated, class-conditional, and geometry-conditioned feature masking—all achieving net ≈ 0 at 25 seeds.

We identify two root causes predictable from geometric statistics: feature saturation for flat/thin objects (flat_frac ≥ 0.4 strongly correlated with regression, $\rho = -0.96$) and orientation ambiguity for asymmetric tools (σ_ψ high, no object-frame encoding). These diagnoses point to three requirements for a successor method: *(i)* object-frame-normalised orientation features or object-aware pose canonicalisation; *(ii)* model-uncertainty-aware gating that disables reranking when score spread falls below a meaningful threshold; *(iii)* explicit handling of degenerate-H regimes for thin planar objects.

Our work demonstrates that improving offline pair accuracy is necessary but not sufficient for online task benefit, and provides a replicable diagnostic framework for predicting, per object class, whether a geometry-only reranker is likely to help or harm.

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